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CASE STUDY · SMALL BUSINESS · INFRA MODERNIZATION

Small Business Infrastructure Modernization
Server 2012 → 2022 + Hyper-V → ESXi (Loaner-Server Pattern)

How Technijian modernized an SMB's Hyper-V server stack onto Windows Server 2022 and VMware ESXi using a loaner-server pattern — zero-purchase transition risk, sustained production, no weekend big-bang.

PROFILE	ENVIRONMENT	SERVICES ENGAGED	ENGAGEMENT SCOPE
Small Business / SMB	**2 Hyper-V hosts, mixed VM fleet**	**Server OS, Hypervisor, Migration**	**84 hrs across 6 phases**

THE CHALLENGE

A small business ran production workloads on **two aging Windows Server 2012 hosts using Hyper-V**. Microsoft had ended mainstream support for the OS. The business needed to move to **Windows Server 2022** and wanted to **standardize on VMware ESXi** as its hypervisor going forward — but couldn't afford to buy a third permanent host just to migrate.

The team also couldn't tolerate a weekend big-bang where both Hyper-V hosts were down together. VMs had to keep running for the sales, accounting, and file-share services the business depended on every weekday.

- ✓ **Two Hyper-V hosts on Server 2012** past end-of-life
- ✓ **Production VMs** supporting business-critical services
- ✓ **Hypervisor shift** — Hyper-V to ESXi for strategic standardization
- ✓ **No budget for a third permanent host** just to enable migration
- ✓ **Zero tolerance** for weekend big-bang dual-host downtime
- ✓ **Load balancing** needed between the two final hosts at the end

THE SOLUTION — LOANER-SERVER PATTERN

Technijian solved the "no third host" problem with a **loaner-server pattern**: install a temporary ESXi loaner onsite, migrate production VMs onto it, then rebuild each original Hyper-V host as ESXi and migrate VMs back. The business never loses both hosts at once, never buys hardware it doesn't need, and ends up on ESXi + Server 2022 with load balancing between the two upgraded hosts.

PHASE 1
Planning & Prep
Thorough assessment of current server environment; detailed migration plan; timelines; resource allocation; risk and mitigation strategies documented.

PHASE 2
Loaner Install
Install and configure the loaner ESXi server onsite. Verify connectivity and performance before any VM touches it.

PHASE 3
Build & Migrate
Build new Windows Server 2022 VMs on the loaner ESXi host. Migrate data and applications from old Hyper-V VMs to the new Server 2022 VMs on loaner.

PHASE 4
HV01 Upgrade
Decommission old servers on HV01. Install ESXi on HV01. Migrate VMs from HV01 to the loaner (keeping production live), then begin preparing HV02 for upgrade.

PHASE 5
HV02 Upgrade
Migrate remaining VMs from HV02 to HV01 (now running ESXi). Upgrade HV02 to ESXi. Migrate VMs back for load balancing.

PHASE 6
Balance & Remove
Balance VMs between HV01 and HV02 for optimal performance. Remove the loaner server from the environment. Document the new architecture.

WHY THE LOANER-SERVER PATTERN BEATS BIG-BANG

- ✓ **No third permanent host** — loaner goes home at the end
- ✓ **Always one production host up** — never dark both at once
- ✓ **Rollback path** at every phase boundary
- ✓ **Business-safe pace** — not weekend pressure

Before: Server 2012 + Hyper-V | After: Server 2022 + ESXi | vCenter management | 2-host load balance

ENGAGEMENT SCALE

6	84	2→2	0
PHASES DELIVERED	HOURS TOTAL	HOSTS (LOANER OUT)	DUAL-HOST OUTAGES

Modernize the stack. Keep the business up. Send the loaner home. 949.379.8500

THE RESULTS

The business ended up on **two modern ESXi hosts** running **Windows Server 2022 VMs**, with load balancing between them and the loaner server retired. Production services stayed up throughout. No hardware was purchased that didn't need to exist. The decommissioned legacy OS and hypervisor are gone from the environment.

OUTCOME METRIC	RESULT
OS modernization	Production workloads moved from Windows Server 2012 onto Windows Server 2022 — back on a supported OS baseline for security and compliance.
Hypervisor standardization	Environment migrated from Hyper-V to VMware ESXi with vCenter management — aligning with the strategic hypervisor direction.
Zero dual-host outage	Because the loaner server carried production during each host upgrade, HV01 and HV02 were never down at the same time.
No unnecessary hardware	Loaner-server pattern eliminated the need to buy a permanent third host just to enable migration — loaner returned at project end.
Load balancing	Final VM placement balanced across the two ESXi hosts for optimal performance — not an accidental leftover distribution.
Legacy decommission	Windows Server 2012 instances and Hyper-V role formally retired; old VMs archived per data-retention requirements before removal.
Rollback discipline	Every phase boundary had a rollback path; at no point was the business in a "have to finish this weekend or we're in trouble" state.

REPRESENTATIVE WORK

Environment assessment — Full inventory of VMs, applications, data, network dependencies, and business-hours sensitivity before any change; plan reviewed and signed off by stakeholders.

Loaner onsite install — ESXi loaner server installed, configured, and validated for connectivity, storage, and performance before any production VM migrated onto it.

Server 2022 VM build — New Windows Server 2022 VMs built on the loaner ESXi host. Applications and data migrated from old Hyper-V VMs to the new VMs.

HV01 rebuild — After production traffic moved to loaner, old VMs on HV01 decommissioned; ESXi installed on HV01; VMs migrated from loaner to HV01.

HV02 rebuild — Remaining VMs moved from HV02 to HV01 (now ESXi); HV02 upgraded to ESXi; VMs balanced back for optimal performance.

Loaner removal & docs — Loaner server decommissioned and returned; final architecture documented; vCenter management configured for ongoing operations.

REPRESENTATIVE OUTCOME

> *"A small business modernized from two Windows Server 2012 / Hyper-V hosts to two Windows Server 2022 / ESXi hosts with load balancing — using a loaner-server pattern that preserved production uptime throughout and avoided purchasing a third permanent host."*
>
> — Anonymized engagement summary · based on 84 hr server-infrastructure modernization proposal

Modernize without weekend panic — or unnecessary hardware spend.

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